

AI Capstone Spring 2026

Course Project Specification

Deadline: 2026/06/07 23:59

Overview

The AI Capstone Course Project is designed to provide students with hands-on experience across the full lifecycle of **Physical AI**—from real-world data collection, to policy learning, and ultimately to evaluation.

To support diverse backgrounds, learning goals, and levels of ambition, the course is structured into three progressive tiers: **Entry**, **Advanced**, and **Independent Study**. Each level features distinct learning activities and assessment criteria, enabling students to engage meaningfully while pursuing increasing levels of technical depth, creativity, and research-oriented exploration.

This tiered design not only builds strong foundational competence but also establishes a clear pathway from structured learning to open-ended innovation and real-world deployment. By tightly integrating simulation, learning, and physical system validation, the course project offers a unified and distinctive framework for training the next generation of Physical AI talents.

Course Project Hierarchy

	ENTRY	ADVANCED	INDEPENDENT STUDY
PARTICIPATION TYPE	Group	Group	Individual
ATTENDANCE	Mandatory	Optional	Optional
GRADE WEIGHTING	80%	+20% on top of Entry level	+30%
SUBMISSION INSTRUCTIONS	- All submission files must be uploaded to the appointed Google Drive. - For more details, please refer to the Submission section.	included in entry submission.	- Please discuss and confirm the appropriate format with your mentor. - Submit individual report separately to the E3 system.

This project consists of both **group and individual components**. The **Entry Level** is a mandatory group project that accounts for 80% of the final grade. The **Advanced Level** is an optional extension of the group work, providing an additional 20% on top of the Entry score. The **Independent Study Level** is an optional individual component that offers up to 30% bonus. In other words, **completing both the Entry and Advanced group components allows you to achieve up to 100% of the final project grade. If you also complete the individual component, you can earn up to 130% in total.** In addition, there is a leaderboard mechanism in the Entry Level: the group with the highest performance in each task will receive an extra 5% bonus score.

Group Project -----

Entry Level (80%)

(Please note that participation at this level is mandatory, and all students must join a group.)

At the Entry level, students will experience the full lifecycle of Physical AI within a predefined tasks, training, and evaluation framework. Given the practical constraints in accessing robotic systems capable of real-world interaction through sensors and actuators, we develop a **Robot Learning framework** that emulates such interactions within a physics-based simulation environment. This framework enables students to experience both **perception and action loops** through embodied interaction between a robotic agent and a simulated environment.

The framework provides:

- a base simulation environment built on NVIDIA Isaac Sim [5]
- a suite of baseline manipulation tasks (e.g., object picking and placing)
- reference policies integrated from the LeRobot ecosystem [4]
- and standardized evaluation metrics (e.g., task success rate)

This unified setup lowers the barrier to entry while preserving the essential characteristics of Physical AI systems, allowing students to focus on understanding core principles such as perception-action loop, policy learning, and policy evaluation.

Each group should select and complete one of three provided tasks (see the [following section](#) for details), on a first-come, first-served basis via a [shared sign-up sheet](#).

Within this level, each group is allowed to explore and modify the following two aspects: training data and policy model. Please note that any changes beyond these two aspects are not allowed:

- **Training data:** including the data itself (i.e., what constitutes good or suitable data), dataset size (i.e., how much data is needed), and data collection methods.
 - At the Entry level, we provide two data collection methods: UMI [1] with motion planning and keyboard control, as detailed in the following

section. Students are expected to explore how much data is needed under limited time and computational resources. To give you hands-on experience with real-world data collection, a time slot will be scheduled for each group to collect data using UMI in ED305.

- **Policy model:** including model architecture and corresponding hyperparameters. You are encouraged to explore different model designs and tuning strategies. The LeRobot ecosystem [4] used in this project can be used as a reference.

Students will participate in a two-stage evaluation process involving Public and Private Leaderboards. During the active development phase, a public validation configuration will be provided to track real-time performance. The final rankings will be determined by a separate private test configuration to ensure model generalization and integrity. Further details regarding the release of both leaderboards will be shared in subsequent announcements.

Grading Policy:

Component	Weight (in final grade)
Technical Report	40%
Presentation Video	40%
The top-performing group in each task	+5% (on top of Entry level)

Advanced Level (+20%, Optional)

(Participation in this level is optional and left to each group's discretion.)

The Advanced level encourages deeper exploration and innovation by allowing students to **define new tasks beyond the Entry level**, including task definition, environment settings, and evaluation metrics. Supporting resources (e.g., 3D assets and reference implementations) are provided in [X-Humanoid](#), [Synthesis](#), and [USD Assets Working Group](#) [6-8] to facilitate creative and technically rigorous project extensions. **Please note that all work at this level is conducted in simulation.**

Students **MUST** complete the Entry level before progressing to the Advanced level. Both levels share a unified development framework, enabling a seamless transition from structured learning to open-ended research and system design.

This level emphasizes:

- Originality and innovation
- Problem formulation
- End-to-end system design and evaluation

Students are encouraged to explicitly address sim-to-real considerations and discuss potential real-world deployment challenges. Projects at this level are expected to approach the scope, rigor, and quality of a workshop-level research contribution.

Grading Policy:

Component	Weight (in final grade)
Technical Report	+10% (on top of Entry level)
Presentation Video	+10% (on top of Entry level)

Reproducibility Requirements:

At this advanced level, students are responsible for defining their own tasks. To ensure external validation, submissions must include a comprehensive implementation guideline. The provided document should outline step-by-step instructions that allow our team to reproduce the claimed results.

- Code submission with clear documentation
- Fixed evaluation protocol (e.g., random seeds, dataset splits)
- Reproducibility checklist to ensure consistent benchmarking

Individual Project -----

Independent Study Level (+ 30%, Optional)

The Independent Study level offers a unique opportunity for students to participate in ongoing **research-oriented projects** in collaboration with members of the HCIS Lab.

Students will have the opportunity to engage in frontier research, collaborate closely with researchers, and gain hands-on experience with advanced robotic systems, experimental platforms, and evaluation. This includes exposure to real-world challenges such as perception uncertainty, system integration, safety considerations, human–robot interaction, and ontology-based data verification.

This level is designed for highly motivated students seeking deeper research engagement and is intended to bridge coursework with cutting-edge academic and real-world applications.

Research Projects

Project Name	Mentor
Knowledge Representation and Formal Ontology for PhyAI	張俊彥
Agentic Mobile Manipulation	林翔恩
Adaptive Policy Learning for Meal Preparation	陳晉祿
ToddlerBot	陳奕廷

Project Name	Mentor
Robotic Food Acquisition for Assistive Feeding	戴熾玲
Real-World Bimanual Manipulation	林谷翰、黃毓翔、鄔仁迪、周士傑
Vision Language Model-enabled UAV Autonomy	曾子昕
Safe and Convenient Object Handover for In-Home Assistive Manipulation	張家睿、蔡茗鈞

Grading Policy:

Component	Weight (in final grade)
Technical Report	+20%
Mentor's Endorsement	+10%

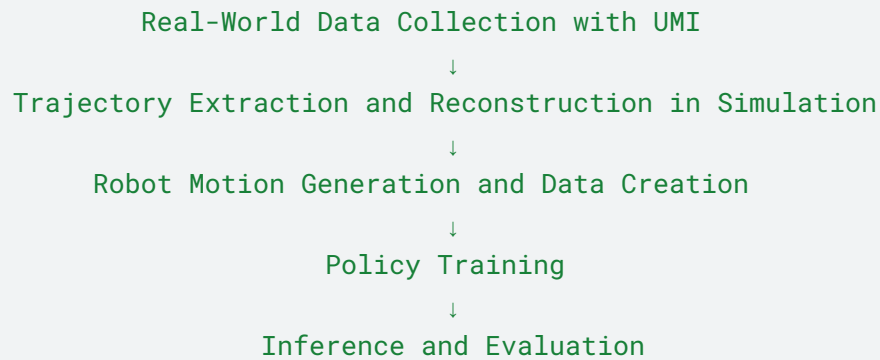
- **Technical Report:** A comprehensive report documenting the project scope, methodology, experimental results, and critical analysis. The report should clearly articulate problem formulation, system design, implementation details, and reflections on real-world deployment challenges.
- **Mentor's Endorsement:**
 - **Process Engagement:** Evaluation of the student's engagement and contribution throughout the research process, including initiative, collaboration with mentors, consistency of progress, and ability to incorporate feedback.
 - **Project-Specific Components:** Additional evaluation components may be incorporated at the discretion of the project mentor, depending on the domain and project scope. These may include system demonstrations, experimental validations, prototype development, or other research-oriented deliverables.

A Robot Learning Framework

In this section, we present a detailed description of the robot learning framework used in this course project for the **Entry level**. An overview of the framework is illustrated below.

Overview

None



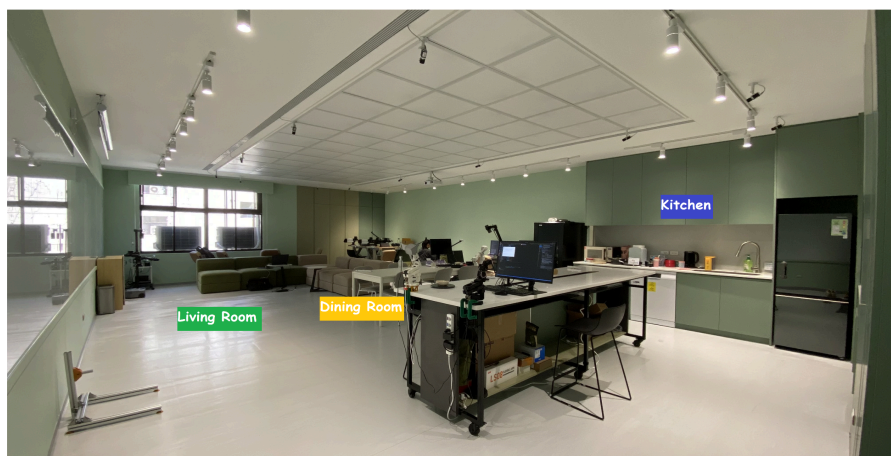
Step 1. Real-World Data Collection with UMI

To provide hands-on experience with real-world data collection, we employ the Universal Manipulation Interface (UMI) [1] as a hand-held device to collect human demonstration data in the ED305 classroom. This process enables students to capture natural, task-relevant behaviors that serve as the foundation for subsequent policy learning.

Each group is assigned one of the following pick-and-place tasks:

- **Cup Stacking (Kitchen Scenario):**
Randomly place a blue cup and a pink cup upside down on the countertop. The task is to pick up the blue cup and stack it on top of the pink cup.
- **Cutlery Arrangement (Dining Room Scenario):**
Randomly place a fork and a knife on the table. The task is to position the knife on the right side of the plate and the fork on the left side.
- **Toy Block Collection (Living Room Scenario):**
Randomly scatter three toy blocks on the tabletop. The task is to collect all scattered blocks and place them into a designated basket.

For more detailed information, please refer to the [UMI Data Collection Guideline](#).



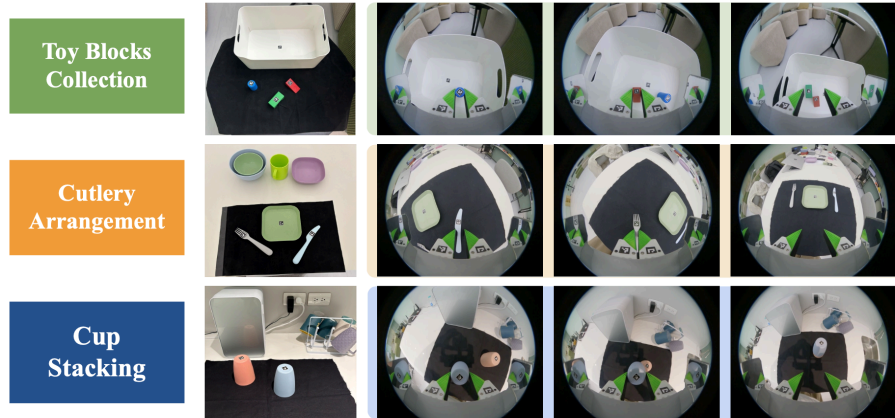


Figure 1. Illustrations of the predefined tasks in the ED305 classroom

Step 2. Reconstruction in Simulation

In this step, students reconstruct real-world demonstrations within NVIDIA Isaac Sim [5] through a provided systematic pipeline. Due to practical constraints, we do not provide real robots at this level. As an alternative, we employ a simulation environment to support extensive testing and development, while improving accessibility, scalability, and experimental reproducibility.

To bridge the gap between real-world data collection and simulation-based experimentation, the collected demonstrations are reconstructed in the simulation environment. In this process, **only object poses from the real-world data are used to set up the simulation scene, while the recorded human trajectories are not directly transferred**. This design choice is due to several practical limitations:

- Noise and inaccuracies in trajectory extraction from demonstrations
- Errors in spatial configuration estimation
- The sim-to-real gap in dynamics and control
- Complex physical interactions and collision handling in simulation
- Variability in embodiment (e.g., differences between human motion and robot kinematics)

Instead of replaying raw human trajectories, a motion planning algorithm is used to generate feasible robot actions in simulation, as discussed in the next section. Transferring real-world human motion trajectories into simulation remains challenging. Students can further explore improvements to this pipeline, and promising solutions will be incorporated into the course repository.

Step 3. Robot Motion Generation and Data Creation

Following the reconstruction process, robot motions are generated using a **finite state machine-based planner** [9, 10]. Given the reconstructed object states and

task configurations, the planner produces feasible action sequences that satisfy kinematic and environmental constraints within the simulation.

In addition to the planner-based pipeline, **keyboard teleoperation** is available as an alternative method for collecting demonstration data in simulation. It can be used to supplement real-world data when it is insufficient, or to support extended data collection in the Advanced level. Students may also implement their own data collection methods as an extension.

To ensure data quality, explicit task success criteria (e.g., correct object placement, completion within constraints) are defined and applied to all collected trajectories. Only successful executions are recorded and stored as valid training data. The resulting dataset is organized in the format defined by the open-sourced LeRobot ecosystem [4], enabling efficient downstream policy training.

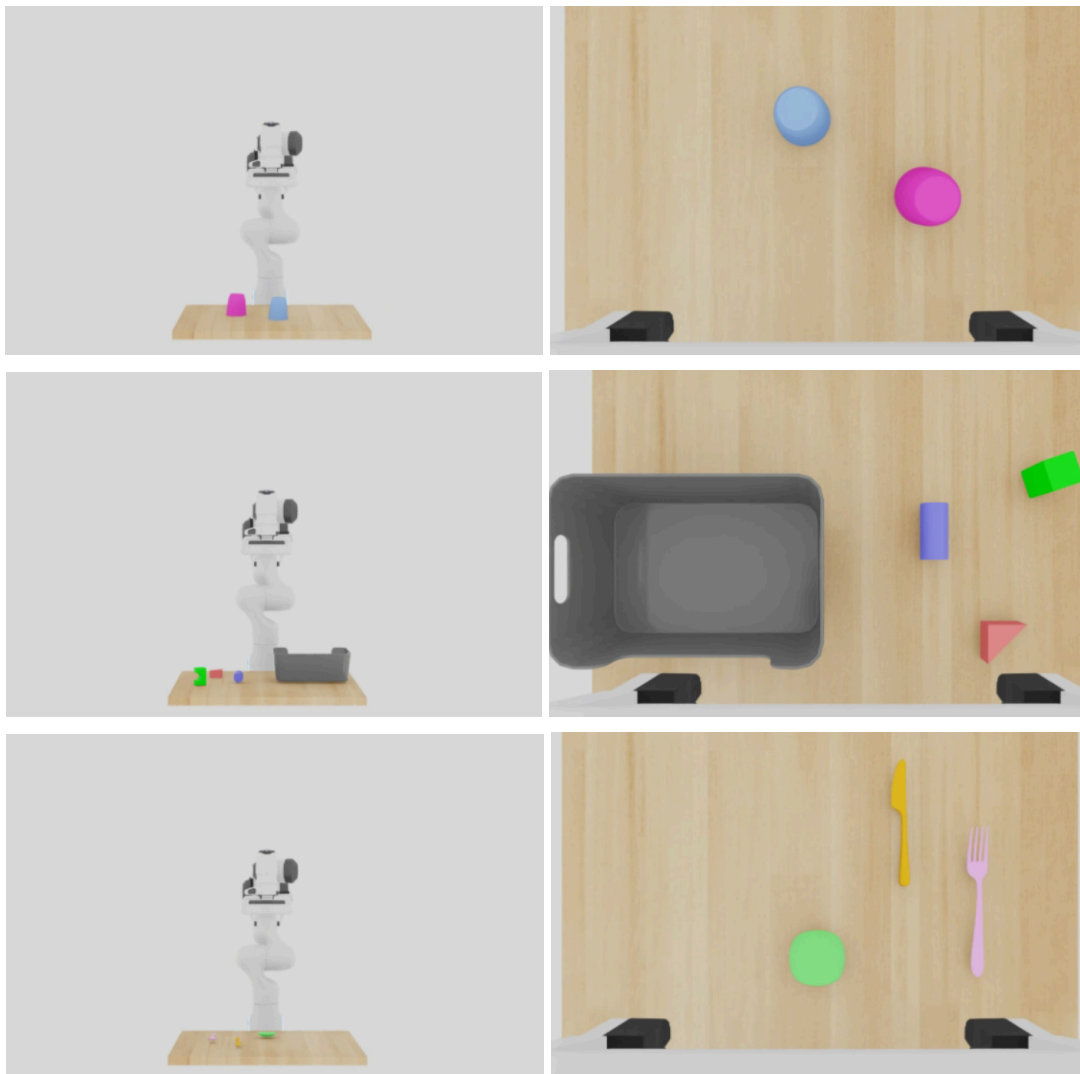


Figure 2. Illustrations of the predefined tasks in the simulation environment, including Cup Stacking (top), Toy Block Collection (middle), and Cutlery Arrangement (bottom).

Extension for Advanced Level

In the Advanced level, students are expected to move beyond the predefined tasks, objects, and environments introduced in the Entry level. Specifically, students will:

- Design new tasks, environments, and object configurations
- Develop customized data collection strategies aligned with their task definitions
- Generate or curate assets (e.g., using USD-based scene descriptions) to support their experimental setup

While existing resources [6-8] could be used as a starting point, students are encouraged to extend beyond them and construct original, well-motivated problem settings. This stage emphasizes end-to-end system thinking, requiring students to jointly consider task design, data generation, and policy learning.

Step 4. Policy Training

Using the dataset generated in the previous step, we train policies to map observations to robot actions. We leverage a suite of robot learning algorithms provided by the **LeRobot ecosystem** [4]. As a starting point, we recommend Diffusion Policy [3], a state-of-the-art imitation learning approach for visuomotor control. Given visual observations and robot states, the policy learns to predict action sequences (i.e., robot trajectories).

Students are encouraged to:

- Explore alternative policy architectures within or beyond the provided framework.
- Tune training strategies (e.g., data augmentation, normalization, horizon length, etc.) Analyze the relationship between dataset quality and policy performance.

This step emphasizes learning-based control, bridging perception and action through a scalable policy learning framework.

Step 5. Inference and Evaluation

After training, policies are evaluated under varying object positions to assess their robustness and generalization capabilities. The evaluation is conducted under two modes: leaderboard and local evaluation.

- **Leaderboard:** We provide public and private leaderboards that only allow N submissions per day. More details will be announced later.
- **Local evaluation:** Rollout the trained policy on your side.

Performance on the private leaderboard will be ranked **separately for each task** based on predefined metrics. The top-performing group in each task will receive a bonus **5 points** toward the final course project score. Further details on evaluation criteria, ranking methodology, and bonus allocation will be provided in a separate document. The rules for the leaderboard may be adjusted on a rolling basis.

Recommendations

To help you get started effectively, consider the following key principles:

1. Data Quality is Critical

Model performance is fundamentally driven by the quality and relevance of the data you collect. While collecting additional real-world data outside your assigned time slot is not permitted, you are encouraged to **improve and extend the data generation pipeline** to increase both the quantity and diversity of training data. Any modifications must be clearly documented in both your report and presentation.

Possible directions include (but are not limited to):

- **Revisiting Real-World UMI Data Processing:** Trajectory extraction from hand-held devices such as UMI is inherently noisy and may not always yield valid demonstrations. You are encouraged to:
 - Analyze the data processing pipeline (Visualize trajectories, inspect failure cases, and compare successful vs. failed samples)
 - Identify sources of failure (e.g., tracking errors, pose estimation noise)
 - Propose improvements to enhance robustness

Improving this stage is particularly important, as the simulation pipeline is highly dependent on real-world data quality.

- **Enhancing Simulation Data Generation:** You may modify the existing simulation pipeline to improve diversity and robustness, for example:
 - Introduce multiple grasp poses or randomized motion strategies
 - Inject controlled randomness in robot motion to improve generalization
- **Designing Custom Data Collection Pipelines:** Implement own data collection interfaces or pipelines to generate additional training data.

2. Hyperparameter Tuning Matters

Model performance is highly sensitive to hyperparameter choices. While you may begin with default training settings, we strongly recommend consulting the original Diffusion Policy [3] paper for deeper insights into effective configurations. In particular, pay close attention to hyperparameters related to **task horizon length**,

temporal consistency, multi-step behaviors, and action representation. Systematic experimentation with different parameter combinations is strongly encouraged, as it can significantly impact policy performance.

Platform and Resources

We have prepared the following resources to support you in completing the Entry level:

- **Monorepo for AI Capstone ([link](#)):** The main codebase for the final project.
- **Universal Manipulation Interface (UMI) ([link](#)):** A data collection and policy learning framework used in this project, enabling direct skill transfer from in-the-wild human demonstrations to deployable robot policies.
 - **UMI Data Collection ([guide](#)):** Instructions for using UMI to collect real-world data in ED305.
- **Final Project QA Page ([link](#)):** Please feel free to ask questions.
- **Glows.AI ([guide](#)):** This platform provides GPU resources. We will assign credits to the team leader after your team has finished collecting real-world data with UMI.

Submission

The deadline for submission of the project is **06/07 (Sun.) at 23:59**. Each group needs to submit the following to the **assigned Google Drive folder**:

- **Presentation Video:** Video named **Team{TEAM_ID}_presentation.mp4**, in which you briefly explain your project work. The video should be **within 15 minutes**.
- **Project Report:** Report named **Team{TEAM_ID}_project_entry_report.pdf**. Groups that have completed the advanced requirements must also submit **Team{TEAM_ID}_project_advanced_report.pdf**.

Both reports should provide a detailed description of your implementation, design choices, and insights developed throughout the project. Please follow the guidelines outlined in the “Report Format” section below.

- **Checkpoint Folder:** A directory containing all trained model checkpoints. Refer to [docs/lerobot-model-format.md](#) for required structure and contents.
- **Configurations Folder:** A directory containing all necessary environment and task configuration files. See [docs/standalone_env_config_export.md](#) for export requirements. **This is required for Advanced level only.**

- **Custom CAD Models:** Submission of all 3D models used in the project, which **must be provided in the .usd file format**. **This is required for the Advanced level only, if custom assets are used.**
- **README.txt:** A mandatory file clearly specifying the complete execution guidelines for your code. For Advanced Level submissions, this file must include a comprehensive implementation guide that enables the evaluation team to independently reproduce all claimed results, explicitly stating if no additional files or custom configurations are needed to ensure benchmarking integrity.
- **Supplementary Files:** Any additional supporting files necessary for reproducibility or complete documentation of the project.

Note:

- **All submission files must be uploaded to the appointed Google Drive folder. A [form](#) has been provided, which includes the **Google Drive link** for each group. Please fill in the email address of one designated group member (team leader) for each group in the form by **May 11**. We will grant Google Drive access permission to that student on May 12 so they can upload the files.**
- Once the submission deadline has passed, no further modifications are allowed. **If any file is found to have a modification timestamp later than the deadline, it will be treated as a late submission and penalized accordingly.**
- If any file within the submitted link cannot be opened or downloaded properly, the part of submission will not be graded.
- For this final project, we **do not accept any late submissions**.
- Any formatting errors will receive a **10-point deduction**.

Presentation Video Format

There is no strict format that you must follow. You are encouraged to organize your presentation in a clear manner that best communicates your work. All presentations must be delivered **in English**. **If you have completed the advanced level, include it in the same video as the entry-level presentation.** The presentation should be **within 15 minutes**, and must address the following key aspects:

- **Task Description:** Clearly describe the assigned task, problem setting, and objectives.
- **Pipeline and Method Modifications:** Explain any modifications or improvements you made to the provided pipeline, models, or codebase.
- **Key Learnings and Insights:** Reflect on what you learned from the project, including challenges encountered, solutions explored, and key takeaways.

A well-structured presentation should emphasize clarity, technical depth, and critical reflection, rather than simply reporting results.

Report Format

All groups must follow the required report structure:

- **Entry-level work:** Complete all sections without the “(Advanced)” label. Sections marked “(Advanced)” must not be included.
- **Advanced work:** Only groups that complete the Advanced requirements should include and complete the sections marked “(Advanced)”.

Group Report Template:

1. Task Description

- Introduce the task you are working on.
- (Advanced) Please describe the motivation of the proposed new task.

2. Data Collection

- **(Entry-Only) Real-World Data Collection with UMI**
 - Explain how you collected the data using the UMI device.
 - Describe how the collected demonstrations were converted into a training dataset.
 - Provide visualizations of the collected data.
- **Simulation Data Collection in Isaac Sim**
 - Describe dataset size and the percentage of each collection methods (UMI with finite state machine–based planner / keyboard teleoperation /self-implemented method)
 - (Optional) If you have self-implemented data collection methods, provide your justifications and implementation details.
 - Provide visualizations of the collected data with the [provided tool](#).
 - (Advanced) Describe your changes and design compared to the Entry level.

3. Policy Training

- **Model Architecture**
 - Describe the model architecture used in your project.
 - (Optional) If you modified the original architecture, explain the changes and your motivations and observations.
- **Training Procedure**
 - Describe the overall training procedure.

- List and explain the key hyperparameters.
- (Optional) If you adopted a different training procedure, describe your approach and the reasons for your design choices.

4. Experimental Results

- **Data Quality Analysis**
 - Explore and identify methods to analyze your collected data, and determine whether it is of sufficient quality and suitable for use in robot learning.
- **Quantitative and Qualitative Result of Policy Models**
 - Figure of training loss curves to illustrate the learning process of your models. Describe and analyze your results, including performance trends and convergence behavior.
 - Provide the calculated success rate over N rollout times on your side, *refer to “#Rollout section” of the readme in the “aicapstone” repository.*
 - Provide screenshots of the rollout procedure of your trained policy.
 - Any you want to share.

(Advanced) Students are expected to provide a detailed description of their performance evaluation process. Please include:

- **Evaluation Procedure:** Clearly describe the evaluation criteria and environment configurations, and explain the rationale behind their design.
- **Results:** Present the outcomes of your evaluation using tables, figures, or numerical results if applicable.

5. Discussion

Share your insights, challenges encountered during the project, and lessons learned. You may also discuss potential improvements or future directions.

6. Work Distribution

- Each team member's assigned tasks and responsibilities.
- Each team member's percentage of work contribution.

Individual Report:

- Please discuss and confirm the report format with your mentor.
- Students who complete an individual project must submit a report to the E3 system under “**Independent Study Report.**”

References

- [1] Chi et al., Universal manipulation interface: In-the-wild robot teaching without in-the-wild robots. RSS 2024. Project page: <https://umi-gripper.github.io/>
- [2] Isaac Sim Document: <https://docs.isaacsim.omniverse.nvidia.com/5.1.0/index.html>
- [3] Chi, et al. Diffusion policy: Visuomotor policy learning via action diffusion. The International Journal of Robotics Research, 44.10-11 (2025): 1684-1704. Project Page: <https://diffusion-policy.cs.columbia.edu/>
- [4] LeRobot: <https://huggingface.co/lerobot>
- [5] Isaac Lab: <https://developer.nvidia.com/isaac/lab>
- [6] X-Humanoid/ArtVIP: [Hugging Face](#)
- [7] Synthesis: <https://synthesis.extwin.com/#/home>
- [8] USD Assets Working Group: <https://github.com/usd-wg/assets/tree/main>
- [9] Finite-state machine: [Finite-state machine Wiki](#)
- [10] Pick and Place implementation: [IsaacSim's Official Implementation](#)
- [11] genrobot2025/10Kh-RealOmin-OpenData: [Hugging Face](#)